

Администрирование локальных сетей

Предварительная настройка оборудования Cisco

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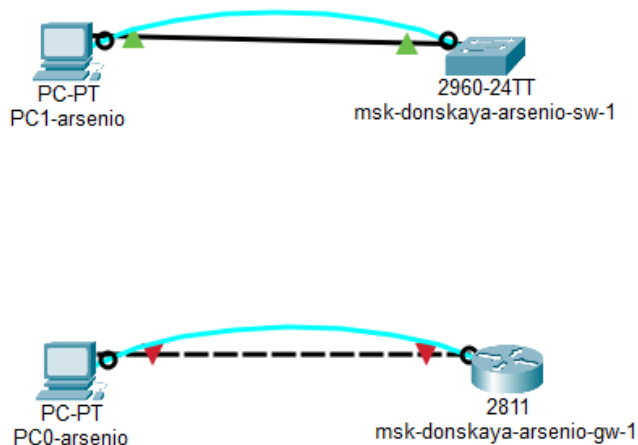
Российский университет дружбы народов, Москва, Россия

Цели и задачи работы

Получить основные навыки начального конфигурирования оборудования Cisco в среде Cisco Packet Tracer.

Построение схемы

Схема подключения оборудования



Настройка маршрутизатора

Конфигурация маршрутизатора

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#host msk-donskaya-arsenio-gw-1
msk-donskaya-arsenio-gw-1(config)#int f0/0
msk-donskaya-arsenio-gw-1(config-if)#no shut

msk-donskaya-arsenio-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

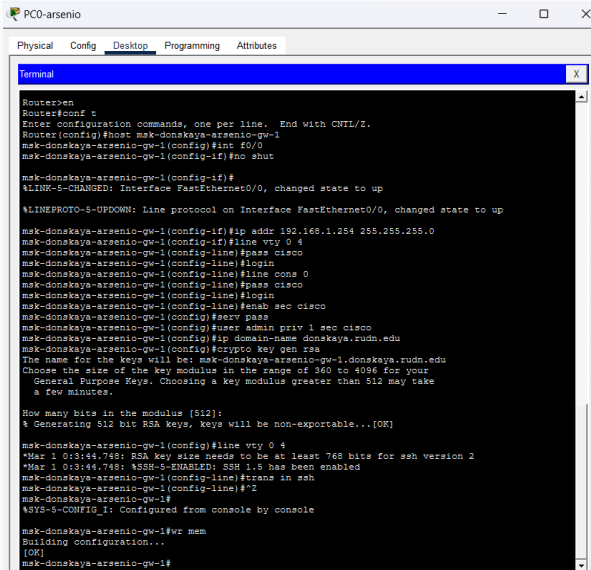
msk-donskaya-arsenio-gw-1(config-if)#ip addr 192.168.1.254 255.255.255.0
msk-donskaya-arsenio-gw-1(config-if)#line vty 0 4
msk-donskaya-arsenio-gw-1(config-line)#pass cisco
msk-donskaya-arsenio-gw-1(config-line)#login
msk-donskaya-arsenio-gw-1(config-line)#line cons 0
msk-donskaya-arsenio-gw-1(config-line)#pass cisco
msk-donskaya-arsenio-gw-1(config-line)#login
msk-donskaya-arsenio-gw-1(config-line)#enab sec cisco
msk-donskaya-arsenio-gw-1(config)#serv pass
msk-donskaya-arsenio-gw-1(config)#user admin priv 1 sec cisco
msk-donskaya-arsenio-gw-1(config)#ip domain-name donsкаya.rudn.edu
msk-donskaya-arsenio-gw-1(config)#crypto key gen rsa
The name for the keys will be: msk-donskaya-arsenio-gw-1.donskaya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-arsenio-gw-1(config)#line vty 0 4
*Mar 1 0:3:44.748: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:3:44.748: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-donskaya-arsenio-gw-1(config-line)#trans in ssh
msk-donskaya-arsenio-gw-1(config-line)#^Z
msk-donskaya-arsenio-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-arsenio-gw-1#wr mem
Building configuration...
[OK]
msk-donskaya-arsenio-gw-1#
```

Консольная настройка маршрутизатора



PC0-arsenio

Physical Config Desktop Programming Attributes

Terminal

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#host msk-donskaya-arsenio-gw-1
msk-donskaya-arsenio-gw-1(config)#int f0/0
msk-donskaya-arsenio-gw-1(config-if)#no shut

msk-donskaya-arsenio-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

msk-donskaya-arsenio-gw-1(config-if)#ip addr 192.168.1.254 255.255.255.0
msk-donskaya-arsenio-gw-1(config-if)#line vty 0 4
msk-donskaya-arsenio-gw-1(config-line)#pass cisco
msk-donskaya-arsenio-gw-1(config-line)#login
msk-donskaya-arsenio-gw-1(config-line)#line cons 0
msk-donskaya-arsenio-gw-1(config-line)#pass cisco
msk-donskaya-arsenio-gw-1(config-line)#login
msk-donskaya-arsenio-gw-1(config-line)#enab sec cisco
msk-donskaya-arsenio-gw-1(config)#serv pass
msk-donskaya-arsenio-gw-1(config)#user admin priv 1 sec cisco
msk-donskaya-arsenio-gw-1(config)#ip domain-name donskeya.rudn.edu
msk-donskaya-arsenio-gw-1(config)#crypto key gen rsa
The name for the keys will be: msk-donskaya-arsenio-gw-1.donskeya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-arsenio-gw-1(config)#line vty 0 4
*Mar 1 0:3:44.748: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:3:44.748: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-donskaya-arsenio-gw-1(config-line)#trans in ssh
msk-donskaya-arsenio-gw-1(config-line)#^2
msk-donskaya-arsenio-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-arsenio-gw-1#wr mem
Building configuration...
[OK]
msk-donskaya-arsenio-gw-1#
```


Настройка коммутатора

Конфигурация коммутатора

```
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#host msk-donskaya-arsenio-sw-1
msk-donskaya-arsenio-sw-1(config)#int vlan 2
msk-donskaya-arsenio-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan2, changed state to up

msk-donskaya-arsenio-sw-1(config-if)#no sh
msk-donskaya-arsenio-sw-1(config-if)#ip addr 192.168.2.1 255.255.255.0
msk-donskaya-arsenio-sw-1(config-if)#int f0/1
msk-donskaya-arsenio-sw-1(config-if)#sw mode acc
msk-donskaya-arsenio-sw-1(config-if)#sw acc vlan 2
msk-donskaya-arsenio-sw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan2, changed state to up

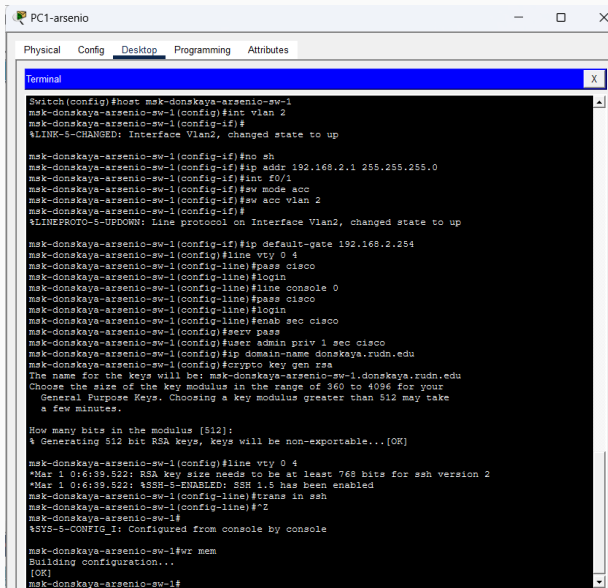
msk-donskaya-arsenio-sw-1(config-if)#ip default-gate 192.168.2.254
msk-donskaya-arsenio-sw-1(config)#line vty 0 4
msk-donskaya-arsenio-sw-1(config-line)#pass cisco
msk-donskaya-arsenio-sw-1(config-line)#login
msk-donskaya-arsenio-sw-1(config-line)#line console 0
msk-donskaya-arsenio-sw-1(config-line)#pass cisco
msk-donskaya-arsenio-sw-1(config-line)#login
msk-donskaya-arsenio-sw-1(config-line)#enab sec cisco
msk-donskaya-arsenio-sw-1(config)#serv pass
msk-donskaya-arsenio-sw-1(config)#user admin priv 1 sec cisco
msk-donskaya-arsenio-sw-1(config)#ip domain-name donsKaya.rudn.edu
msk-donskaya-arsenio-sw-1(config)#crypto key gen rsa
The name for the keys will be: msk-donskaya-arsenio-sw-1.donskaya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-arsenio-sw-1(config)#line vty 0 4
*Mar 1 0:6:39.522: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:6:39.522: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-donskaya-arsenio-sw-1(config-line)#trans in ssh
msk-donskaya-arsenio-sw-1(config-line)#^Z
msk-donskaya-arsenio-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-arsenio-sw-1#wr mem
Building configuration...
[OK]
msk-donskaya-arsenio-sw-1#
```

Консольная настройка коммутатора



PC1-arsenio

Physical Config Desktop Programming Attributes

Terminal

```
Switch(config)#host msk-donskaya-arsenio-sw-1
msk-donskaya-arsenio-sw-1(config)#int vlan 2
msk-donskaya-arsenio-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan2, changed state to up

msk-donskaya-arsenio-sw-1(config-if)#no sh
msk-donskaya-arsenio-sw-1(config-if)#ip addr 192.168.2.1 255.255.255.0
msk-donskaya-arsenio-sw-1(config-if)#int f0/1
msk-donskaya-arsenio-sw-1(config-if)#sw mode acc
msk-donskaya-arsenio-sw-1(config-if)#sw acc vlan 2
msk-donskaya-arsenio-sw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan2, changed state to up

msk-donskaya-arsenio-sw-1(config-if)#ip default-gate 192.168.2.254
msk-donskaya-arsenio-sw-1(config)#line vty 0 4
msk-donskaya-arsenio-sw-1(config-line)#pass cisco
msk-donskaya-arsenio-sw-1(config-line)#login
msk-donskaya-arsenio-sw-1(config-line)#line console 0
msk-donskaya-arsenio-sw-1(config-line)#pass cisco
msk-donskaya-arsenio-sw-1(config-line)#login
msk-donskaya-arsenio-sw-1(config-line)#enab sec cisco
msk-donskaya-arsenio-sw-1(config)#serv pass
msk-donskaya-arsenio-sw-1(config)#user admin priv 1 sec cisco
msk-donskaya-arsenio-sw-1(config)#ip domain-name donsкаya.rudn.edu
msk-donskaya-arsenio-sw-1(config)#crypto key gen rsa
The name for the keys will be: msk-donskaya-arsenio-sw-1.donsкаya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-arsenio-sw-1(config)#line vty 0 4
*Mar 1 0:6:39.522: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:6:39.522: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-donskaya-arsenio-sw-1(config-line)#trans in ssh
msk-donskaya-arsenio-sw-1(config-line)#^Z
msk-donskaya-arsenio-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-arsenio-sw-1#wr mem
Building configuration...
[OK]
msk-donskaya-arsenio-sw-1#
```

Настройка оконечных устройств

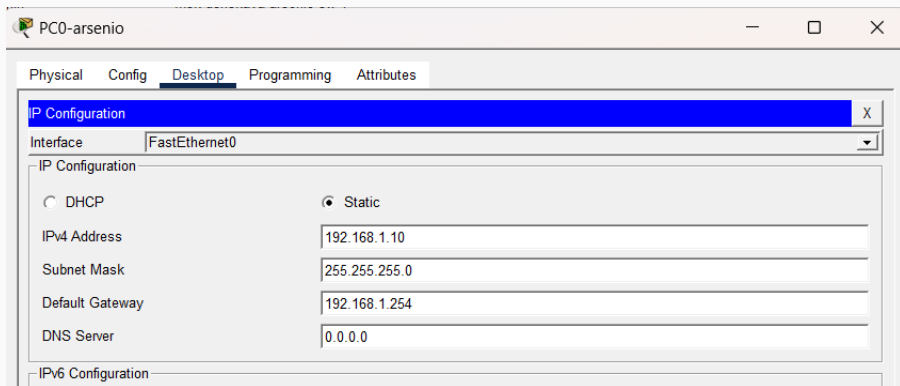


Рис. 6: Настройка IP-адреса на PC0-arsenio

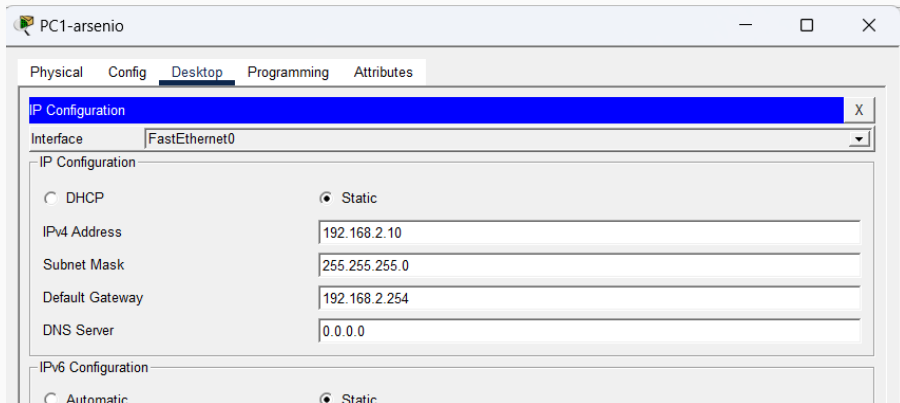
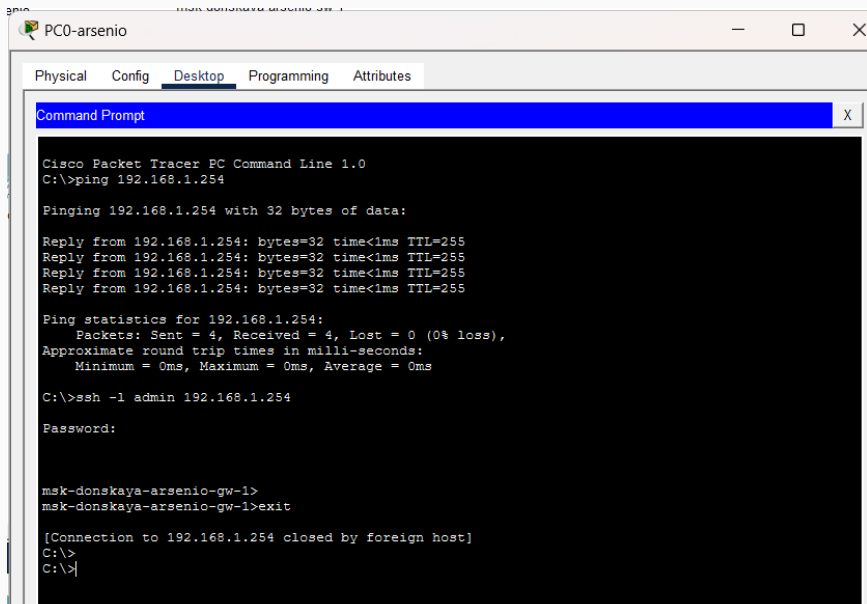


Рис. 7: Настройка IP-адреса на PC1-arsenio

Проверка соединений

Проверка маршрутизатора



```
PC0-arsenio
Physical Config Desktop Programming Attributes
Command Prompt X
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.254

Pinging 192.168.1.254 with 32 bytes of data:

Reply from 192.168.1.254: bytes=32 time<1ms TTL=255
Reply from 192.168.1.254: bytes=32 time<1ms TTL=255
Reply from 192.168.1.254: bytes=32 time<1ms TTL=255
Reply from 192.168.1.254: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.254:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

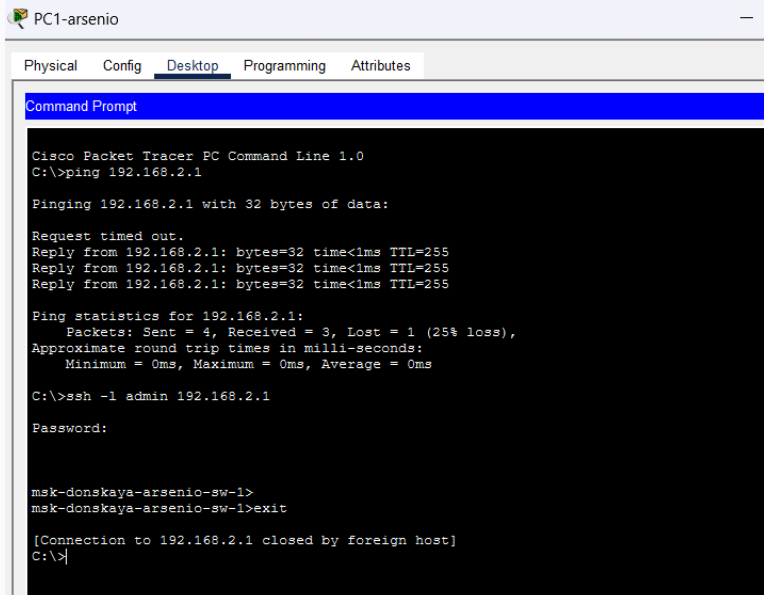
C:\>ssh -l admin 192.168.1.254

Password:

msk-donskaya-arsenio-gw-1>
msk-donskaya-arsenio-gw-1>exit

[Connection to 192.168.1.254 closed by foreign host]
C:\>
C:\>|
```


Проверка коммутатора



The screenshot shows a Cisco Packet Tracer PC Command Line window for a device named 'PC1-arsenio'. The window has tabs for 'Physical', 'Config', 'Desktop', 'Programming', and 'Attributes', with 'Desktop' currently selected. The command prompt shows the execution of a ping command to 192.168.2.1, which results in a 25% packet loss. Subsequently, an SSH command is entered to connect to the same IP address as the 'admin' user, but the connection is closed by the foreign host.

```
PC1-arsenio
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.1

Pinging 192.168.2.1 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.2.1:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ssh -l admin 192.168.2.1

Password:

msk-donskaya-arsenio-sw-1>
msk-donskaya-arsenio-sw-1>exit

[Connection to 192.168.2.1 closed by foreign host]
C:\>
```

Выводы

В ходе лабораторной работы была выполнена предварительная настройка маршрутизатора Cisco **2811** и коммутатора **2960-24TT**.

Были настроены IP-адреса, имена устройств, пароли, локальный пользователь и SSH-доступ.

Проверка командой **ping** подтвердила доступность маршрутизатора и коммутатора.

Подключение по **SSH** показало, что удалённое администрирование настроено корректно.